about the distance he has fallen and the status of his electronic systems, much as a radio station encodes a newscast on the radio waves it transmits.

Back in the starship your crew will receive the laser beam, and Kares will decode it to get the distance and system information. She will also measure the beam's wavelength (or, equivalently, its color; see Figure P.2). The wavelength is important; it tells how fast Arnold is moving. As he moves faster and faster away from the starship, the green beam he transmits gets *Doppler-shifted*,[19] as received at the ship, to longer and longer wavelengths; that is, it gets more and more red. (There is an additional shift to the. red caused by the beam's struggle against the hole's gravitational pull. When computing Arnold's speed, Kares must correct her calculations for this *gravitational redshift*[20])

And so the experiment begins. Arnold blasts his way out of orbit and onto an infalling trajectory. As he begins to fall, Kares starts a clock to time the arrival of his laser signals. When 10 seconds have elapsed, the decoded laser signal reports that all his systems are functioning well, and that he has already fallen a distance of 2630 kilometers. From the color of the laser light, Kares computes that he is now moving inward with a speed of 530 kilometers per second. When the ticking clock has reached 20 seconds his speed has doubled to 1060 kilometers per second and his distance of fall has quadrupled to 10,500 kilometers. The clock ticks on. At 60 seconds his speed is 9700 kilometers per second, and he has fallen 135,000 kilometers, five-sixths of the way to the horizon.

You now must pay very close attention. The next few seconds will be crucial, so Kares turns on a high-speed recording system to collect all details of the incoming data. At 61 seconds Arnold reports all systems still functioning normally; the horizon is 14,000 kilometers below him and he is falling toward it at 13,000 kilometers per second. At 61.7 seconds all is still well, 1700 kilometers more to go, speed 39,000 kilometers per second, or about one-tenth the speed of light, laser color beginning to change rapidly. In the next one-tenth of one second you watch in amazement as the laser color zooms through the electromagnetic spectrum, from green to red, to infrared, to microwave, to radio-wave, to—. By 61.8 seconds it is all over. The laser beam is completely gone. Arnold has reached the speed of light and disappeared into the horizon. And in that last tenth of a second, just